E2 Notes: complex numbers

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1. Why complex numbers?

The usual story is "$x^2 = -1$ has no solution, so we invent one". That is not how it happened, and the real story is more convincing.

In the 1540s Cardano published a formula for cubic equations. For $x^3 = px + q$ it gives

$$x = \sqrt[3]{\frac{q}{2} + \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}} + \sqrt[3]{\frac{q}{2} - \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}}.$$

Around 1572 Bombelli applied it to $x^3 = 15x + 4$, which obviously has the solution $x = 4$. The formula gives

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}.$$

Square roots of negative numbers appear in the middle of the calculation even though the answer is a perfectly good real number. Bombelli noticed that if you calculate with $\sqrt{-1}$ using the ordinary rules, then $(2 + \sqrt{-1})^3 = 2 + 11\sqrt{-1}$ (check it!), so the cube roots are $2 \pm \sqrt{-1}$ and their sum is 4. Complex numbers were forced on mathematicians as a tool for finding *real* answers. Problem 16 asks you to redo Bombelli's calculation.

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2. Construction

How do we know that "calculating with $\sqrt{-1}$" does not lead to a contradiction? We build a number system in which it is safe.

Definition. A complex number is an ordered pair (a, b) of real numbers, with

$$(a, b) + (c, d) = (a + c,\ b + d), \qquad (a, b) \cdot (c, d) = (ac - bd,\ ad + bc).$$

The set of all such pairs is $\mathbb{C}$.

- Pairs $(a, 0)$ behave exactly like real numbers: $(a, 0) + (c, 0) = (a + c, 0)$ and $(a, 0)(c, 0) = (ac, 0)$. So we write a for $(a, 0)$ and see $\mathbb{R}$ as part of $\mathbb{C}$.
- Put $i = (0, 1)$. Then $i \cdot i = (0 \cdot 0 - 1 \cdot 1,\ 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1$. **So $i^2 = -1$ is a theorem.**
- Every pair can be written $(a, b) = (a, 0) + (b, 0)(0, 1) = a + bi$.

One can check (tedious but straightforward) that addition and multiplication are commutative and associative, that multiplication distributes over addition, that 0 and 1 are neutral elements, and that every nonzero element has an inverse (see §3). In the language of algebra: $\mathbb{C}$ is a **field** (*lichaam*), just like $\mathbb{Q}$ and $\mathbb{R}$. All the usual algebra (expanding brackets, the formula $(a + b)^2 = a^2 + 2ab + b^2$, the factor theorem) therefore works in $\mathbb{C}$.

What we lose: order. There is no way to define "$<$" on $\mathbb{C}$ with the usual rules (if $x > 0$ and $y > 0$ then $xy > 0$; for $x \neq 0$ either $x > 0$ or $-x > 0$). *Proof:* $i \neq 0$, so $i > 0$ or $-i > 0$. In both

cases squaring gives $i^2 = (-i)^2 = -1 > 0$. Then $1 = (-1)^2 > 0$ too, so $0 = 1 + (-1) > 0$, a contradiction. So never write $z < w$ for non-real complex numbers.

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3. Arithmetic

For $z = a + bi$ with $a, b \in \mathbb{R}$: the **real part** $\operatorname{Re} z = a$, the **imaginary part** $\operatorname{Im} z = b$ (a real number, without the i!), and the **conjugate** $\bar{z} = a - bi$.

Rules for the conjugate. For all z, w:

$$\overline{z + w} = \bar{z} + \bar{w}, \qquad \overline{zw} = \bar{z}\,\bar{w}, \qquad z + \bar{z} = 2\operatorname{Re} z, \qquad z\bar{z} = a^2 + b^2.$$

Proof of the second: with $z = a + bi$, $w = c + di$: $zw = (ac - bd) + (ad + bc)i$, so $\overline{zw} = (ac - bd) - (ad + bc)i$, and $\bar{z}\bar{w} = (a - bi)(c - di) = (ac - bd) + (-ad - bc)i$. Equal. ■

Division. Since $z\bar{z} = a^2 + b^2$ is a positive real number when $z \neq 0$, multiply numerator and denominator by the conjugate of the denominator:

$$\frac{2 - i}{1 + 3i} = \frac{(2 - i)(1 - 3i)}{(1 + 3i)(1 - 3i)} = \frac{2 - 6i - i + 3i^2}{10} = \frac{-1 - 7i}{10}.$$

In particular $\dfrac{1}{z} = \dfrac{\bar{z}}{a^2 + b^2}$.

Powers of i. $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, then it repeats with period 4. So $i^{2031} = i^{4 \cdot 507 + 3} = -i$.

Equality. $a + bi = c + di$ (with a, b, c, d real) iff $a = c$ and $b = d$. This is used to split one complex equation into two real ones. Example: find $\sqrt{5 - 12i}$. Put $(a + bi)^2 = 5 - 12i$: $a^2 - b^2 = 5$ and $2ab = -12$. Also $|\cdot|^2$ gives $a^2 + b^2 = 13$. So $a^2 = 9$, $b^2 = 4$, and $ab < 0$: $\pm(3 - 2i)$.

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4. The complex plane

Draw $z = a + bi$ as the point (a, b), or the vector from 0 to (a, b). The horizontal axis is the **real axis** (*reële as*), the vertical the **imaginary axis** (*imaginaire as*).

- Addition $z + w$ is vector addition (parallelogram rule).
- $\bar{z}$ is the reflection in the real axis; $-z$ is the point reflection in 0.
- The **modulus** $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$ is the distance from 0 to z, and $|z - w|$ is the distance between z and w.

Properties of the modulus.

1. $|zw| = |z|\,|w|$. *Proof:* $|zw|^2 = zw\overline{zw} = z\bar{z}\,w\bar{w} = |z|^2|w|^2$. ■
2. **Triangle inequality** $|z + w| \leq |z| + |w|$. *Proof:*

$$|z + w|^2 = (z + w)(\bar{z} + \bar{w}) = |z|^2 + |w|^2 + z\bar{w} + \bar{z}w = |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w}).$$

Since $\operatorname{Re}(u) \leq |u|$ for every u, and $|z\bar{w}| = |z||w|$, this is at most $(|z| + |w|)^2$. ■

Sets in the plane. Many conditions on z describe familiar figures:

Condition	Set
$\lvert z-c\rvert = r$	circle with centre c, radius r
$\lvert z-c\rvert < r$	open disc
$\lvert z-a\rvert = \lvert z-b\rvert$	perpendicular bisector (*middelloodlijn*) of a and b
$\arg(z-c) = \theta$	half-line from c (not including c) at angle θ
$\lvert z-a\rvert = k\lvert z-b\rvert$, $k \neq 1$	a circle (circle of Apollonius)

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5. Polar form

A nonzero z is determined by its modulus $r = |z|$ and the angle φ from the positive real axis to the vector z, the **argument** $\arg z$. Then

$$z = r(\cos\varphi + i\sin\varphi).$$

The argument is defined only up to multiples of 2π. The **principal argument** (*hoofdwaarde*) is the one in $(-\pi, \pi]$.

Finding the argument. $\tan\varphi = b/a$, but arctan only gives angles in $(-\frac{\pi}{2}, \frac{\pi}{2})$. Always draw z and choose the angle in the right quadrant. Example: $z = -3 - 3i$ lies in the third quadrant; $\tan\varphi = 1$ and the right angle is $\varphi = -\frac{3\pi}{4}$ (not $\frac{\pi}{4}$). And $|z| = 3\sqrt{2}$.

Theorem (multiplication). If $z = r(\cos\alpha + i\sin\alpha)$ and $w = s(\cos\beta + i\sin\beta)$, then

$$zw = rs\left(\cos(\alpha+\beta) + i\sin(\alpha+\beta)\right).$$

Proof. Expand:

$$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta) = (\cos\alpha\cos\beta - \sin\alpha\sin\beta) + i(\sin\alpha\cos\beta + \cos\alpha\sin\beta),$$

and by the addition formulas this is $\cos(\alpha+\beta) + i\sin(\alpha+\beta)$. ∎

Multiply the moduli, add the arguments. So multiplying by w is a **rotation** by $\arg w$ combined with a **scaling** by $|w|$. Multiplying by i rotates by $90°$. Similarly $\dfrac{z}{w}$ has modulus r/s and argument $\alpha - \beta$.

This theorem is the real reason complex numbers matter: they turn rotations into multiplications.

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6. De Moivre's theorem

Theorem (De Moivre). For every integer n,

$$(\cos\varphi + i\sin\varphi)^n = \cos n\varphi + i\sin n\varphi.$$

Proof for $n \geq 0$ by induction. For $n = 0$ both sides are 1. If it holds for n, then by the multiplication theorem

$$(\cos\varphi + i\sin\varphi)^{n+1} = (\cos n\varphi + i\sin n\varphi)(\cos\varphi + i\sin\varphi) = \cos(n+1)\varphi + i\sin(n+1)\varphi.$$

For negative $n = -m$: $(\cos\varphi + i\sin\varphi)^{-1} = \cos\varphi - i\sin\varphi = \cos(-\varphi) + i\sin(-\varphi)$, then apply the case $m \geq 0$. ■

Application 1: powers. $(1+i)^{10}$: $1 + i = \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4})$, so $(1+i)^{10} = 32(\cos\frac{10\pi}{4} + i\sin\frac{10\pi}{4}) = 32(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) = 32i$.

Application 2: multiple-angle formulas. Write $c = \cos\varphi$, $s = \sin\varphi$. By De Moivre and the binomial theorem,

$$\cos 3\varphi + i\sin 3\varphi = (c + is)^3 = c^3 + 3ic^2s - 3cs^2 - is^3.$$

Comparing real and imaginary parts:

$$\cos 3\varphi = c^3 - 3cs^2 = 4c^3 - 3c, \qquad \sin 3\varphi = 3c^2s - s^3 = 3s - 4s^3.$$

No need to memorise these: derive them in a minute when needed.

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7. Euler's formula

Define, for real φ,

$$e^{i\varphi} = \cos\varphi + i\sin\varphi.$$

Why use the letter e? Because this function behaves exactly like an exponential:

1. **Exponent rule.** $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$: that is the multiplication theorem of §5. And De Moivre becomes $(e^{i\varphi})^n = e^{in\varphi}$.
2. **Derivative.** Differentiating real and imaginary parts separately, $\dfrac{d}{d\varphi}(\cos\varphi + i\sin\varphi) = -\sin\varphi + i\cos\varphi = i(\cos\varphi + i\sin\varphi)$. So $f(\varphi) = e^{i\varphi}$ satisfies $f' = if$ with $f(0) = 1$, just as $e^{k\varphi}$ satisfies $f' = kf$, $f(0) = 1$ (compare E1 §6).
3. **Power series** (going further). Substituting $x = i\varphi$ in $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$ and splitting into even and odd powers gives exactly the series of $\cos\varphi$ plus i times that of $\sin\varphi$.

For $z = a + bi$ we then define $e^z = e^a e^{ib}$, and $e^{z+w} = e^z e^w$ holds for all complex z, w.

Consequences.

- **Euler's identity:** $e^{i\pi} = -1$, i.e. $e^{i\pi} + 1 = 0$.
- **Polar form, short:** $z = re^{i\varphi}$.
- **Cosine and sine as exponentials:** adding and subtracting $e^{i\varphi}$ and $e^{-i\varphi} = \cos\varphi - i\sin\varphi$:

$$\cos\varphi = \frac{e^{i\varphi} + e^{-i\varphi}}{2}, \qquad \sin\varphi = \frac{e^{i\varphi} - e^{-i\varphi}}{2i}.$$

These turn trigonometry into algebra. Example: $\cos^2\varphi = \frac{1}{4}(e^{2i\varphi} + 2 + e^{-2i\varphi}) = \frac{1}{2} + \frac{1}{2}\cos 2\varphi$.

- **Integrals.** $\int e^x \cos x\,dx = \operatorname{Re}\int e^{(1+i)x}\,dx = \operatorname{Re}\dfrac{e^{(1+i)x}}{1+i} + C = \tfrac{1}{2}e^x(\cos x + \sin x) + C$ (problem 20). No integration by parts twice.

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8. Roots

Solving $z^n = w$. Write $w = Re^{i\theta}$ ($R > 0$) and look for $z = re^{i\varphi}$. Then $z^n = r^n e^{in\varphi}$, so we need $r^n = R$ and $n\varphi = \theta + 2k\pi$:

$$z_k = \sqrt[n]{R}\, e^{i(\theta+2k\pi)/n}, \qquad k = 0, 1, \dots, n-1.$$

These are n different numbers (for $k = n$ you get z_0 again). They lie on a circle of radius $\sqrt[n]{R}$ and form a **regular n-gon**.

Example. $z^3 = 8i = 8e^{i\pi/2}$: $z = 2e^{i(\pi/6+2k\pi/3)}$, i.e. $2e^{i\pi/6} = \sqrt{3} + i$, $2e^{5i\pi/6} = -\sqrt{3} + i$ and $2e^{3i\pi/2} = -2i$.

Roots of unity. The solutions of $z^n = 1$ are $1, \omega, \omega^2, \dots, \omega^{n-1}$ with $\omega = e^{2\pi i/n}$. For $n \geq 2$ their sum is 0:

$$1 + \omega + \cdots + \omega^{n-1} = \frac{\omega^n - 1}{\omega - 1} = 0,$$

(geometric sum, $\omega \neq 1$). Geometrically: the vectors to the vertices of a regular polygon centred at 0 add up to zero, by symmetry. Taking real parts with $n = 5$ gives $1 + 2\cos\frac{2\pi}{5} + 2\cos\frac{4\pi}{5} = 0$ (problem 14).

Careful with $\sqrt{\ }$. For complex numbers $\sqrt{zw} = \sqrt{z}\sqrt{w}$ fails. The classic false "proof" $-1 = i \cdot i = \sqrt{-1}\sqrt{-1} = \sqrt{(-1)(-1)} = \sqrt{1} = 1$ breaks at the third step. Write "the solutions of $z^2 = w$" instead of $\sqrt{w}$ unless w is a positive real number.

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9. Polynomial equations

Quadratics. The quadratic formula works for complex coefficients too, with "$\sqrt{D}$" meaning either solution of $z^2 = D$. For real coefficients and $D < 0$: $z^2 - 4z + 13 = 0$ gives $z = \frac{4 \pm \sqrt{-36}}{2} = 2 \pm 3i$.

Fundamental theorem of algebra (*hoofdstelling van de algebra*). Every polynomial of degree $n \geq 1$ with complex coefficients has a complex root, and hence (by repeatedly splitting off factors $z - z_1$) factors completely as $a(z - z_1)\cdots(z - z_n)$. The proof needs analysis and is beyond this module (it is in the parent's complex analysis module and is a great goal for later). Accept it for now.

Conjugate root theorem. If P has *real* coefficients and $P(z_0) = 0$, then $P(\bar{z}_0) = 0$.

Proof. $P(z) = a_n z^n + \cdots + a_0$ with all a_k real, so $\bar{a}_k = a_k$. Using the conjugate rules,

$$P(\bar{z}_0) = \sum a_k \bar{z}_0^k = \sum \overline{a_k z_0^k} = \overline{P(z_0)} = \bar{0} = 0. \qquad \blacksquare$$

So non-real roots of real polynomials come in conjugate pairs, and $(z-z_0)(z-\bar{z}_0) = z^2 - 2\mathrm{Re}(z_0)z + |z_0|^2$ is a real quadratic. Consequence: **every real polynomial factors into real linear and real quadratic factors**, and a real polynomial of odd degree has a real root.

Example. $P(z) = z^3-3z^2+7z-5$. Try $z=1$: $1-3+7-5=0$. Divide: $P(z) = (z-1)(z^2-2z+5)$, so the roots are 1 and $1 \pm 2i$.

Cubics: Cardano's method. Every cubic ax^3+bx^2+cx+d can be reduced, by dividing by a and substituting $x = y - \frac{b}{3a}$, to a **depressed cubic** $y^3+py+q=0$ (no square term). Now put $y = u+v$:

$$(u+v)^3 = u^3+v^3+3uv(u+v), \quad \text{so} \quad y^3 - 3uv\,y - (u^3+v^3) = 0.$$

This matches $y^3+py+q=0$ if $uv = -\frac{p}{3}$ and $u^3+v^3 = -q$. Then u^3 and v^3 have sum $-q$ and product $-\frac{p^3}{27}$, so they are the roots of the quadratic

$$t^2 + qt - \frac{p^3}{27} = 0.$$

Example. $x^3+6x-20=0$: $p=6$, $q=-20$. The quadratic $t^2-20t-8=0$ has roots $t = 10 \pm 6\sqrt{3}$. So $x = \sqrt[3]{10+6\sqrt{3}} + \sqrt[3]{10-6\sqrt{3}}$. In fact $(1+\sqrt{3})^3 = 10+6\sqrt{3}$, so $x = (1+\sqrt{3})+(1-\sqrt{3}) = 2$. Dividing by $x-2$ leaves $x^2+2x+10$ with roots $-1 \pm 3i$.

To get all three roots directly from Cardano: if u is one cube root of u^3, the others are $u\omega$ and $u\omega^2$ with $\omega = e^{2\pi i/3}$, and v must be chosen so that $uv = -\frac{p}{3}$. This gives $y = u+v$, $u\omega + v\omega^2$, $u\omega^2 + v\omega$.

The "irreducible case". When the cubic has three distinct real roots, $\frac{q^2}{4} + \frac{p^3}{27} < 0$ and Cardano's formula *necessarily* passes through non-real numbers (Bombelli's situation). In that case the trigonometric method is faster: use $\cos 3\varphi = 4\cos^3\varphi - 3\cos\varphi$ (problem 11).

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10. Geometric transformations as complex functions

Transformation	Complex function
translation over b	$z \mapsto z+b$
rotation about 0 by θ	$z \mapsto e^{i\theta}z$
scaling (*vermenigvuldiging*) from 0 with factor $k>0$	$z \mapsto kz$
rotation about c by θ	$z \mapsto c + e^{i\theta}(z-c)$
reflection in the real axis	$z \mapsto \bar{z}$
reflection in the line through 0 at angle α	$z \mapsto e^{2i\alpha}\bar{z}$

Theorem. Every map $f(z) = az+b$ with $a \neq 0, 1$ is a *spiral similarity* (rotation plus scaling) about its unique fixed point $c = \frac{b}{1-a}$: indeed $f(z) - c = a(z-c)$. (Compare E1: $u_{n+1} = au_n + b$ has fixed point $\frac{b}{1-a}$. Same algebra!)

Example. $f(z) = (1+i)z+2$. Fixed point $c = \frac{2}{1-(1+i)} = \frac{2}{-i} = 2i$. Since $1+i = \sqrt{2}e^{i\pi/4}$, f is a rotation by $45°$ about $2i$ combined with scaling by $\sqrt{2}$ from $2i$.

Proving geometry with complex numbers. Points $0, z, w$ form an equilateral triangle (counterclockwise) iff $w = e^{i\pi/3}z$. With $\rho = e^{i\pi/3}$ we have $\rho^2 - \rho + 1 = 0$ (because $\rho^3 = -1$ and $\rho \neq -1$), so $w^2 - zw + z^2 = z^2(\rho^2 - \rho + 1) = 0$. Problem 22 turns this into a symmetric criterion for any triangle.

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11. Common mistakes

1. **Wrong quadrant for the argument.** $\arg(-1 - i) \neq \frac{\pi}{4}$. Draw the point first.
2. $\mathrm{Im}(a + bi) = bi$**.** No: it is b, a real number.
3. **Dividing without the conjugate**, or multiplying only the numerator by it.
4. **Forgetting roots.** $z^n = w$ has n solutions; listing one is not "solving".
5. **Using** $\sqrt{zw} = \sqrt{z}\sqrt{w}$ for complex (or negative) numbers.
6. **Assuming the conjugate root theorem for non-real coefficients.** $z^2 - iz = 0$ has roots 0 and i, but not $-i$.
7. **Writing inequalities between complex numbers.** Only moduli, real parts, etc. can be compared.
8. **De Moivre with non-integer powers.** $(e^{i\varphi})^{1/2} = e^{i\varphi/2}$ is one choice of square root, not "the" one; that is exactly why roots are multi-valued.

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12. Going further

- i^i **is real.** $i = e^{i\pi/2}$, so "$i^i = e^{i \cdot i\pi/2} = e^{-\pi/2} \approx 0.208$". But $i = e^{i(\pi/2 + 2k\pi)}$ for every integer k, so i^i has infinitely many values $e^{-\pi/2 - 2k\pi}$. Complex powers and logarithms are multi-valued: the start of complex analysis.
- **Möbius transformations** $z \mapsto \frac{az+b}{cz+d}$ map circles and lines to circles and lines. Needham, *Visual Complex Analysis*, chapter 3.
- **The Mandelbrot set**: the c for which $z_{n+1} = z_n^2 + c$, $z_0 = 0$ stays bounded. A complex version of E1's recurrences; program it.
- **Quaternions**: Hamilton (1843) found a 4-dimensional number system $a + bi + cj + dk$ with $i^2 = j^2 = k^2 = ijk = -1$, used today for 3D rotations in computer games and spacecraft. Multiplication is no longer commutative.
- **Constructibility.** The regular 17-gon can be constructed with ruler and compass (Gauss, aged 19) because $\cos\frac{2\pi}{17}$ can be written with square roots only; this comes from studying the 17th roots of unity. A good *profielwerkstuk* topic (see stage F).
- **University connection.** The parent's track has a full complex analysis module.